

Equine Oracle: Updated R&D and Business Strategy Report (Post-MVP)

Date: November 12, 2025 **Author:** Manus AI

1. Executive Summary: Strategic Pivot to Sequential Modeling

The Equine Oracle project has successfully transitioned from a basic MVP to a **sophisticated, production-ready platform** with a clear, aggressive R&D roadmap. The core strategic shift is the move from a simple binary classification model (Top 3 finish) to a **LightGBM Ranker** and the proposal to become a **Sequential Event Modeler** (predicting a streak of winners).

This report integrates the latest technical and business documentation to provide a consolidated strategy for **Equine Oracle Limited** and **Lowkey Consultants Limited**.

2. Core Intelligence Layer: Model Enhancement

The predictive core has undergone a significant upgrade, solidifying the technical advantage for funding applications.

2.1. Model Performance Upgrade (LightGBM Ranker)

The model has been retrained on a massive dataset (78,639 races, 396,572 records) and now uses the industry-standard **LightGBM Ranker** ¹.

Metric	Score	Interpretation	Strategic Value
NDCG@3	0.8579	High accuracy in predicting the top three finishers (Trifecta).	Replaces the less impressive ROC-AUC (0.7893) in all funding documents.
NDCG@4	0.8752	Excellent performance in predicting the top four finishers (First Four).	Quantifiable proof of R&D capability for the Callaghan Grant.

2.2. Feature Proposal: Sequential Event Modeling

The **Oracle Engine Feature Proposal** 2 outlines the next R&D sprint: to become a Sequential Event Modeler capable of predicting a streak of winners (e.g., four consecutive races). This is achieved by integrating high-impact features:

- **Sequential Performance:** `PREV_RACE_WON` , `WIN_STREAK` (capturing horse momentum).
- **Market Efficiency:** `IMPLIED_PROBABILITY` , `NORMALIZED_VOLUME` , `MARKET_ACTIVITY_WINDOW_HOURS` (leveraging the "wisdom of the crowd" from betting markets).

This feature is the **key differentiator** for the B2C subscription model and justifies the premium pricing.

3. Application Architecture and Integration

The MVP's architecture has been enhanced to support the new ML and data requirements.

3.1. Ensemble Prediction System Integration

The `ensemblePredictionService.ts` 3 confirms the robust integration strategy:

- **Hybrid Execution:** The Node.js server uses `execSync` to call a Python script (`run_ensemble.py`) to execute the LightGBM Ranker. This is the correct approach for integrating Python-based ML models into a Node.js environment.
- **Comprehensive Logging:** The service includes detailed logging of every prediction, capturing success/failure, execution time, and model used. This is vital for the **Optimization and Debugging** phase of the R&D workflow.

3.2. Data Synchronization and Quality

New files introduce a critical data quality and ingestion layer:

- **Data Sync Router (`sync_router.ts`):** Provides tRPC procedures (`syncToday` , `syncTomorrow`) for manual or scheduled ingestion of live race data, likely from the TAB API.
- **Data Validator (`data_validator.ts`):** Implements rigorous checks for data quality, including validation of required fields, runner counts, and consistency checks across syncs. This directly addresses the **Risk Mitigation** strategy outlined in the Board Meeting Agenda.

4. Business Strategy and Governance

The **First Board Meeting Agenda** 4 and **Comprehensive R&D Workflow** 5 provide a clear, integrated roadmap for the business.

4.1. Governance and Funding

- **Legal Structure:** Confirms the dual-entity strategy: **Equine Oracle Limited** (IP Holding/Product) and **Lowkey Consultants Limited** (Consulting/R&D).
- **Funding Strategy:** The NDCG@3 score of **0.8579** must be immediately integrated into the **Callaghan Grant** application and the **Angel Investor Pitch Deck** to maximize funding success.
- **Financial Projections:** The \$2M+ EBITDA by Year 3 is contingent on the successful execution of the R&D workflow and the launch of the premium subscription tier.

4.2. R&D Workflow (Build-Measure-Learn)

The R&D workflow is formalized around the **Build-Measure-Learn (BML) loop** 5 .

Phase	Focus	Key Activity	Output
Phase 4	Launch & Data Collection	Deploy MVP, integrate analytics.	Analytics Dashboard, Qualitative Feedback Log.
Phase 6	Optimization & Debugging	Prioritize bug fixes, performance optimization.	MTTR Reduction, Performance Audit Report.
Phase 7	Next Iteration	Hypothesis Refinement (Sequential Modeling).	MVP 2.0 Feature Backlog (Sequential Features).

The next immediate step is to execute **Phase 2: Integrate the enhanced ensemble prediction system and data synchronization features into the MVP** to prepare for the launch.

References

[1] Updated Horse Race Ranking Model Performance Summary.pdf
[2] oracleenginefeature_proposal.pdf
[3] ensemblePredictionService.ts
[4] firstboardmeeting_agenda.pdf
[5] ComprehensiveR&DWorkflowandExecutionPlanfor_.pdf